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AN OPTIMIZATION APPROACH TO RECONSTRUCTING FORCE-FREE FIELDS

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ABSTRACT

A new method for reconstructing force-free magnetic fields from their boundary values, based on minimizing the global departure of an initial field from a force-free and solenoidal state, is presented. The method is tested by application to a known nonlinear solution. We discuss the obstacles to be overcome in the application of this method to the solar case: the reconstruction of force-free fields in the corona from measurements of the vector magnetic field in the low atmosphere.

Subject headings: MHD — Sun: magnetic fields

1. INTRODUCTION

For many situations of interest in the solar corona, forces associated with the magnetic field are expected to dominate over gravity and plasma pressure forces. Stationary states of the magnetic field configuration are then described by the requirement that the Lorentz force be zero,

$$(\nabla \times \boldsymbol{B}) \times \boldsymbol{B} = 0 \,, \tag{1}$$

together with the solenoidal condition,

$$\nabla \cdot \boldsymbol{B} = 0 \,. \tag{2}$$

A magnetic field satisfying equations (1) and (2) is said to be "force free" because the self-force (Lorentz force) is zero. Force-free fields are also relevant to other areas of physics, including hydrodynamics and superconductivity (Marsh 1996).

Equation (1) is basically a nonlinear equation and is remarkably difficult to solve. Some insight into this equation is gained by writing it in the form

$$\nabla \times \boldsymbol{B} = \alpha \boldsymbol{B} \,, \tag{3}$$

where $\alpha = \alpha(\boldsymbol{x})$ is in general a function of position. Taking the divergence of equation (3) gives

$$\boldsymbol{B} \cdot \nabla \alpha = 0 \,, \tag{4}$$

which implies that the value of α is constant along a field line. A special class of force-free solutions that are linear is obtained when α is a constant everywhere. However, these solutions have limited application in the solar case because they do not become potential (current free) at large distances (Sturrock 1994) and because they do not match observed boundary conditions in active regions (e.g., Wang 1993; Pevtsov, Canfield, & Metcalf 1994).

A considerable amount of attention has been devoted to numerical methods of solution of the full nonlinear problem in the solar context. Modern vector magnetograms and Stokes Polarimeter data permit the vector magnetic field in the lower atmosphere above active regions to be inferred from polarization measurements, subject to large errors (Klimchuk & Canfield 1994) and an intrinsic ambiguity in the sign of the transverse field (e.g., Metcalf 1994). These measurements may allow the reconstruction of the field in the corona from its lower boundary values. An important question concerns the appropriate boundary values to ensure a well-posed boundary value problem (e.g., Amari et al. 1997). The solar problem is often approximated as that of determining a force-free field in a half-space; i.e., the effects of the finite field of view of a magnetogram and of the influence of the solar curvature are ignored. In this case it is generally accepted that the appropriate boundary conditions consist of the prescription of the normal component of the magnetic field at the photosphere, together with the value of α for one polarity of the normal field (Aly 1989; Amari et al. 1997; McClymont, Jiao, & Mikic 1997). The well-posedness of the solution is particularly important because of the large errors in the measured boundary values and because the fields are not exactly force free at the photospheric level (e.g., Metcalf et al. 1995). A number of methods have been proposed for solving the force-free equations using the available boundary conditions (for recent reviews, see McClymont et al. 1997; Amari et al. 1997). Reconstructions have been used to investigate questions including the relationship of soft X-ray loops to sites of large vertical current (Metcalf et al. 1994) and the observed thickness of coronal soft X-ray loops (McClymont & Mikic 1994). Potential and linear force-free reconstructions have been used to examine the topology of active region fields and to compare topologically significant sites with the locations of flare emissions (e.g., van Driel-Gesztelyi et al. 1994). Nonlinear methods would be much better suited to this task since the connectivity of the field depends on the spatial distribution of currents (Hudson & Wheatland 1999). It is fair to say that nonlinear force-free field reconstruction methods are in their infancy, and there are many prospective applications of nonlinear methods to the physics of flares and of coronal heating. Granted this, it is worthwhile to investigate new methods of solution of the nonlinear problem.

In this paper we present one such method, involving an optimization procedure. The method is described in § 2, and a test of an implementation of the method, applied to a known (more directly calculable) nonlinear force-free field, is given in § 3. In § 4 we discuss the prospects for the application of this method to solar data.

2. THE METHOD

Consider a vector field $\boldsymbol{B}(\boldsymbol{x}, t)$ defined in a volume V. The method involves the minimization of the quantity

$$L = \int_V [B^{-2} |(\nabla \times \boldsymbol{B}) \times \boldsymbol{B}|^2 + |\nabla \cdot \boldsymbol{B}|^2] dV \tag{5}$$

by an evolutionary procedure. It is clear from the form of equation (5) that, if L is reduced to zero, then $(\nabla \times \boldsymbol{B}) \times \boldsymbol{B} = \nabla \cdot \boldsymbol{B} = 0$ everywhere in V. Hence, it is expected that, if L is reduced to zero, a force-free field exists in the volume V.

Differentiating equation (5) with respect to t and applying a number of vector identities (see Appendix) leads to

$$\frac{1}{2}\frac{dL}{dt} = -\int_V \frac{\partial \boldsymbol{B}}{\partial t} \cdot \boldsymbol{F} dV - \int_S \frac{\partial \boldsymbol{B}}{\partial t} \cdot \boldsymbol{G} dS , \tag{6}$$

where

$$\boldsymbol{F} = \nabla \times (\boldsymbol{\Omega} \times \boldsymbol{B}) - \boldsymbol{\Omega} \times (\nabla \times \boldsymbol{B}) - \nabla(\boldsymbol{\Omega} \cdot \boldsymbol{B}) + \boldsymbol{\Omega}(\nabla \cdot \boldsymbol{B}) + \Omega^2 \boldsymbol{B} \tag{7}$$

and

$$\boldsymbol{G} = \hat{\boldsymbol{n}} \times (\boldsymbol{\Omega} \times \boldsymbol{B}) - \hat{\boldsymbol{n}}(\boldsymbol{\Omega} \cdot \boldsymbol{B}) , \tag{8}$$

with

$$\boldsymbol{\Omega} = B^{-2}[(\nabla \times \boldsymbol{B}) \times \boldsymbol{B} - (\nabla \cdot \boldsymbol{B})\boldsymbol{B}] . \tag{9}$$

The vector $\hat{\boldsymbol{n}}$ is the inward unit normal to S, the surface bounding the volume V. The form of equation (6) implies that L will decrease if $\boldsymbol{B}$ is evolved according to

$$\frac{\partial \boldsymbol{B}}{\partial t} = \mu \boldsymbol{F} , \tag{10}$$

where $\mu > 0$ is an arbitrary function, and if we impose the boundary condition

$$\frac{\partial \boldsymbol{B}}{\partial t} = 0 \tag{11}$$

on S. The evolution of L is then governed by

$$\frac{dL}{dt} = -2 \int_V \mu F^2 dV . \tag{12}$$

Equation (10) provides an explicit prescription for evolving the field so that L decreases. We have not managed to demonstrate that the only possible steady state of that evolution corresponds to a force-free state in the volume V. In the next section, however, we apply the method to a simple example and demonstrate that a good approximation to a force-free field is recovered.

3. TESTING THE METHOD

To test the method a known (calculable) nonlinear solution is used. Low & Lou (1990) have presented a class of axisymmetric nonlinear force-free fields with a multipolar character. The solutions have singular behavior at the origin, but by burying the origin beneath the photosphere and by inclining the axis of symmetry of the fields to the normal, this model provides simple nonlinear fields that mimic coronal fields. The authors suggested that these fields are ideal candidates for testing methods for reconstructing force-free fields from boundary values. The solution used is labeled $P_{1,1}$, with $l = 0.3$ and $\Phi = \pi/4$, in Low & Lou's notation. The calculation of this field involves the solution of a second-order ordinary differential equation, and for this step a shooting method was used (Press et al. 1992).

The method was tested by attempting to reconstruct $P_{1,1}$ from all of its boundary values on the box defined by $-1 < x < 1$, $-1 < y < 1$, and $0 < z < 1$. A modestly sized grid ($40^2 \times 20$) was used to represent fields within the box, with a nonuniform spacing (more grid points were chosen closer to the origin, to better represent the variation in the field near the origin). The first step in the reconstruction of the field was the determination of a potential field within the box to match the vertical component of the field $P_{1,1}$ on the lower boundary. This potential field provided the starting point for the evolution of the field within the box. The correct boundary values for all components of the field $P_{1,1}$ were provided on the lower, side, and top boundaries of the box and were fixed throughout the subsequent calculation, in accordance with the boundary condition (11). The field within the box was evolved by the forward time stepping of equation (10), with the choice $\mu = 1$, for almost 4×10^4 time steps. A time step $\Delta t = 10^{-5}$ was chosen by a process of trial and error (this point is discussed in more detail below). The units of this time step are arbitrary: the test field was normalized to have a maximum value of unity in the lower boundary, and the time step depends on this choice and also on the choice of μ and of the grid spacing.

The left-hand panel of Figure 1 shows the variation of L in the calculation, as a function of the number of time steps. The quantity L is shown to decrease to less than 10^{-4} of its initial value. The calculation was halted when L was still decreasing, albeit very slowly. The measure of success of the method is whether the force-free solution has been recovered, and in the right-hand panel of Figure 1, one indicator of this success is shown. We consider the following current-weighted average of the sine of the angle between the current density and the magnetic field:

$$\sigma_J = \frac{\sum_i J_i \sigma_i}{\sum_i J_i} , \tag{13}$$

where

$$\sigma_i = \sin \theta_i = \frac{|\boldsymbol{J} \times \boldsymbol{B}|_i}{J_i B_i} \tag{14}$$

and where the subscripts i run over grid points within the computational box. In the right-hand panel of Figure 1, $\theta_J = \sin^{-1} \sigma_J$ is shown by the solid curve, and this quantity is seen to decrease to a final value of $\theta_J \approx 0.^\circ 91$. For comparison, the average angle over all grid points, $\langle \theta_i \rangle$, is also shown in the figure, by the dashed curve. The upper panel in Figure 2 is a scatter plot of the individual angles θ_i versus the current density at each point, for the final field. It is clear that points with large current densities have small angles θ_i, although points where the current is weak may have large angles. This plot indicates that the final state of the field is close to force free.

Second, we consider whether the solenoidal condition has been met. The method presented above is unusual among magnetic field calculation methods in that it does not

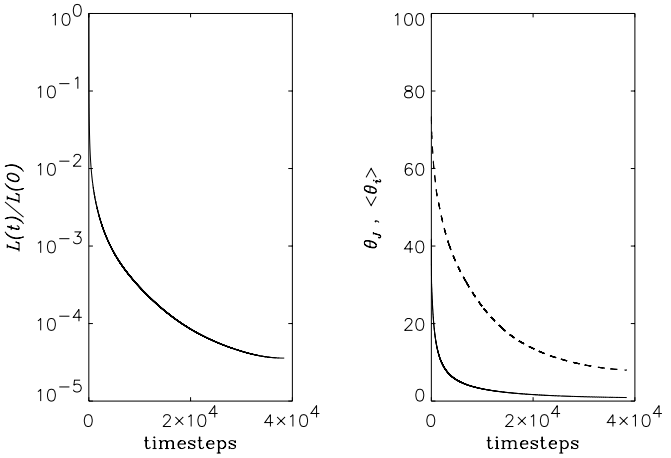


FIG. 1.—*Left:* Time history of the quantity L for the Low & Lou field calculation. *Right:* Time history of the current-weighted average angle between $\boldsymbol{B}$ and $\boldsymbol{J}$ (*solid line*) and of the average angle (*dashed line*).

impose $\nabla \cdot \boldsymbol{B} = 0$ by fiat (for example, by working with a vector potential). Consequently it is important to check that the final state of the system is close to divergence free. Consider the fractional flux increase, f_i, in a small discrete volume ΔV_i (with surface ΔS_i) about the grid point i:

$$f_i = \frac{\int_{\Delta S_i} \boldsymbol{B} \cdot d\boldsymbol{S}}{\int_{\Delta S_i} |\boldsymbol{B}|\, dS} \approx \frac{(\nabla \cdot \boldsymbol{B})_i \Delta V_i}{B_i A_i}, \tag{15}$$

where A_i is the surface area of the small volume. The lower panel in Figure 2 shows the distribution of f_i for our reconstruction. The average value of the magnitude of f_i is $\langle |f_i| \rangle \approx 4.8 \times 10^{-5}$. Figure 2 indicates that the method has achieved a final state that is close to divergence free everywhere in the computational volume.

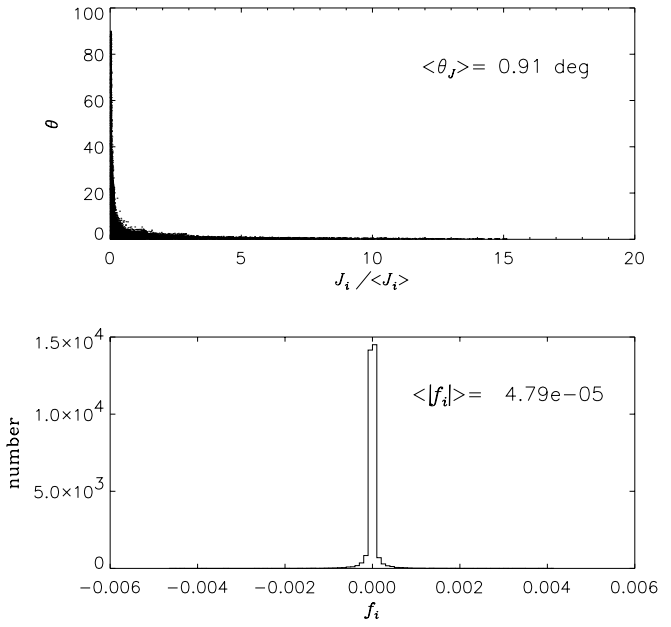


FIG. 2.—*Top:* Scatter plot of angle between current density and magnetic field vs. current density for the Low & Lou field calculation. *Bottom:* Distribution of fractional flux error over all grid points.

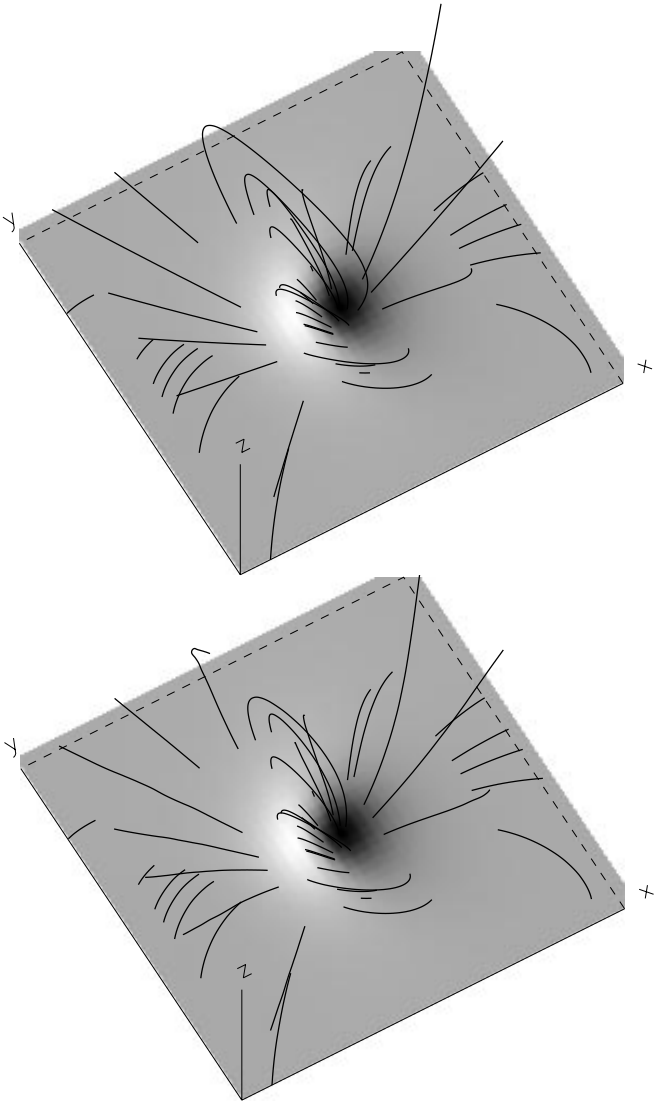


FIG. 3.—*Top:* Field lines calculated for the Low & Lou field. *Bottom:* Field lines for the reconstructed field.

In Figure 3 field line tracings are shown for the "exact" field (*upper panel*) and for the reconstructed field (*lower panel*). In both cases the boundary values of B_z in the lower boundary are shown as a gray-scale image. A small number of field lines, with footpoints equally spaced in the x- and y-directions, are traced for both fields. Most of the field lines are very similar for the original and reconstructed fields. A few of the longer field lines and a few field lines that go near the edge of the computational box (where the grid spacing increases) are appreciably different. It is clear that the general appearance of the field lines has been faithfully reproduced.

4. DISCUSSION

A new method has been presented for reconstructing nonlinear force-free fields from their boundary values, based on an optimization procedure. The method has been tested by application to a calculable nonlinear force-free field and has been shown to be able to recover a good approximation to the original field from the boundary

values of the field. In the following discussion, the advantages and shortcomings of the method are presented, as well as the problems to be overcome in the application of the method to the boundary value problem presented by vector magnetic field measurements on the Sun.

The reconstructed solution obtained in § 3 is only approximately force free and divergence free, as illustrated in Figure 2. The errors in the solution are believed to arise mostly from discretization errors. The accuracy of the prescription for the evolution of the field, equation (10), depends on the accuracy of the representation of the derivatives in the function $\boldsymbol{F}$, and discretization limits this accuracy. A number of numerical experiments support this belief. First, when the method is applied to boundary data for a uniform field, it is found that a uniform field is recovered exactly, to the limits of numerical precision (the gradients in a uniform field can be represented exactly). Second, for the Low & Lou (1990) solution, it is found that increasing the grid resolution or reducing the size of the box in x, y, and z (so that the field within the box varies more smoothly) results in improved measures of success (θ_J, $\langle |f_i| \rangle$) for the reconstructed field.

There are a number of ways in which the implementation of the method could be improved. First, explicit forward time stepping has been used to solve equation (10). While this has the advantage of simplicity, it has the disadvantage that a very small time step is required to ensure numerical stability (e.g., Press et al. 1992). If the time step is too large, the numerical solution will not follow the true evolution of equation (10). In the present context this behavior is easy to detect: it leads to L not decreasing in a time step, which is corrected for by choosing a smaller time step. An appropriate time step is hence found by a process of trial and error. A rough guide to an appropriate choice is that a stable time step Δt is expected to scale with the grid size Δx according to $\Delta t \propto (\Delta x)^2$ (Zwillinger 1989). This estimate of the suitable time step allows a determination of the order of the method, as follows. The requirement that $\Delta t \propto (\Delta x)^2$ implies $\Delta t \propto N^{-2}$, where N is the size of one side of the numerical box. Suppose there is a fixed time (time here refers to the t appearing in eq. [10]) T to convergence. Then the number of time steps required is $T/\Delta t \propto N^2$, and each time step takes CPU time $\propto N^3$ (each point in the volume must be evolved), resulting in a total CPU time $\propto N^5$. Hence the method, as implemented, is expected to be an N^5 process, and numerical experiments roughly confirm this estimate of the order. Such a high-order process makes the method too slow for large boxes. However, this estimate assumes the use of forward time stepping and a fixed, (uniform) spatial mesh of points. It is likely that the speed of the method can be improved by using an implicit evolutionary scheme (which allows larger time steps) and by using a nonuniform grid together with an adaptive grid size. The calculation presented in § 3 uses a (fixed) nonuniform grid, and this was found to improve the speed. The relatively slow speed of the method is basically attributable to the calculation being a three-dimensional evolutionary procedure.

The motivation for developing this method is that of reconstructing force-free fields from the boundary values provided by vector magnetic field measurements in solar active regions. There are a number of obstacles still to be overcome before the method can usefully be applied to real data. First and most important are questions concerning boundary values. In the calculation given in § 2, the boundary values for all components of the magnetic field were prescribed over the surfaces of the bounding box. As discussed in the introduction, the appropriate boundary values for a well-posed problem are expected to be the normal component of the field on each boundary, together with the value of α over one polarity of the normal field. Specifying all components of the field over the bounding surfaces is an overprescription, and if measured solar boundary values are used, errors in the inferred values of α over the two polarities of the normal field may lead to inconsistent boundary conditions. The method requires the boundary condition (11). If (11) is used, then all components of the field must be fixed on all boundaries while the field within the box is being evolved. It is difficult to say how serious a problem the overprescription of boundary values is for the application of the method. This effect will be investigated by adding noise to the Low & Lou solution and repeating the reconstruction presented in this paper. If the problem proves to be serious, then it may be necessary to use an iterative scheme in which the values of α over one polarity of the boundary field are adjusted to match those of the other polarity, in between applications of the method described in this paper. Another problem is that boundary values at the side and top surfaces of the bounding box are required, whereas in practice only the lower boundary values are available. In the absence of correct values for the field on the side and top boundaries of the box, it might be possible to use approximate values, e.g., potential or linear force-free extrapolations of the lower boundary values. The effect of approximate boundary values will also be investigated in the near future via numerical simulations. It is expected that approximate boundary values will lead to inaccuracies in the reconstructed field, but it is unclear how serious the errors will be. The influence of the outer boundaries may be reduced by using a nonuniform grid, with the grid spacing increasing toward the side and top boundaries. Another possibility is that a weighting function can be introduced into the integral in equation (5):

$$L = \int_V w(x, y, z)[B^{-2} |(\nabla \times \boldsymbol{B}) \times \boldsymbol{B}|^2 + |\nabla \cdot \boldsymbol{B}|^2] dV \, , \tag{16}$$

where $w(x, y, z)$ is chosen to go to zero at the side and top boundaries. In this way the method is explicitly directed to ignore departures of the field from the force-free (and solenoidal) state close to the problematic boundaries. It is straightforward to derive the counterparts of equations (6)–(9) for equation (16).

Another question to be addressed in the application of the method to real data sets is the influence of the 180° ambiguity in the azimuthal angle of the measured transverse field. A variety of methods that attempt to resolve this ambiguity using simple physical assumptions exist (e.g., Metcalf 1994), but these techniques are not foolproof. In the near future the effect of the ambiguity on the optimization approach will be examined by flipping some of the azimuthal angles in the Low & Lou (1990) boundary values and repeating the optimization calculation.

Finally, there are a number of other practical concerns. In its present form the code is too slow to be applied to the rather large data sets (typically, $N \sim 200$) provided by modern vector magnetographs and Stokes Polarimeters. As discussed above, changes in the implementation of the

method may greatly improve its speed. Also, the method must be modified to deal with non-Cartesian grids, which are needed to describe coordinate systems defined by magnetic field measurements made away from Sun center.

Provided solutions can be found to the various problems outlined above, the optimization approach may provide a powerful tool for reconstructing magnetic fields in the solar corona. It is also possible that the method will prove useful in other areas where force-free fields play a role.

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APPENDIX

DERIVATION OF EQUATIONS (6)–(9)

Equation (5) may be rewritten

$$L = \int_V B^2 \Omega^2 \, dV \, , \tag{A1}$$

where $\boldsymbol{\Omega}$ is defined by equation (9). Differentiating equation (A1) with respect to t leads to

$$\frac{1}{2}\frac{dL}{dt} = \int_V \left\{ \boldsymbol{\Omega} \cdot \frac{\partial}{\partial t} [(\nabla \times \boldsymbol{B}) \times \boldsymbol{B} - (\nabla \cdot \boldsymbol{B})\boldsymbol{B}] - \Omega^2 \boldsymbol{B} \cdot \frac{\partial \boldsymbol{B}}{\partial t} \right\} dV \, . \tag{A2}$$

The goal is to rewrite the right-hand side of equation (A1) as surface and volume integrals involving a dot product with $\partial \boldsymbol{B}/\partial t$ (see eq. [6]). The second term under the integral sign in equation (A2) is already in the required form. The first term is not and requires some manipulation, beginning with the expansion

$$H = \int_V \boldsymbol{\Omega} \cdot \left[\left(\nabla \times \frac{\partial \boldsymbol{B}}{\partial t} \right) \times \boldsymbol{B} + (\nabla \times \boldsymbol{B}) \times \frac{\partial \boldsymbol{B}}{\partial t} - \left(\nabla \cdot \frac{\partial \boldsymbol{B}}{\partial t} \right) \boldsymbol{B} - \frac{\partial \boldsymbol{B}}{\partial t} \nabla \cdot \boldsymbol{B} \right] dV \, . \tag{A3}$$

The first term in (A3) may be rewritten:

$$\begin{aligned} H_1 &= \int_V \boldsymbol{\Omega} \cdot \left[\left(\nabla \times \frac{\partial \boldsymbol{B}}{\partial t} \right) \times \boldsymbol{B} \right] dV \\ &= \int_V \left(\nabla \times \frac{\partial \boldsymbol{B}}{\partial t} \right) \cdot (\boldsymbol{B} \times \boldsymbol{\Omega}) dV \\ &= \int_V \left\{ \frac{\partial \boldsymbol{B}}{\partial t} \cdot [\nabla \times (\boldsymbol{B} \times \boldsymbol{\Omega})] - \nabla \cdot \left[(\boldsymbol{B} \times \boldsymbol{\Omega}) \times \frac{\partial \boldsymbol{B}}{\partial t} \right] \right\} dV \, , \end{aligned} \tag{A4}$$

using vector identities. Next Gauss's law is applied to the second term under the integral sign in equation (A4),

$$\begin{aligned} H_1 &= \int_V \frac{\partial \boldsymbol{B}}{\partial t} \cdot [\nabla \times (\boldsymbol{B} \times \boldsymbol{\Omega})] dV + \int_S \hat{\boldsymbol{n}} \cdot \left[(\boldsymbol{B} \times \boldsymbol{\Omega}) \times \frac{\partial \boldsymbol{B}}{\partial t} \right] dS \\ &= \int_V \frac{\partial \boldsymbol{B}}{\partial t} \cdot [\nabla \times (\boldsymbol{B} \times \boldsymbol{\Omega})] dV + \int_S \frac{\partial \boldsymbol{B}}{\partial t} \cdot [\hat{\boldsymbol{n}} \times (\boldsymbol{B} \times \boldsymbol{\Omega})] dS \, , \end{aligned} \tag{A5}$$

to obtain the required form.

The second term in equation (A3) may be brought into the required form using a vector identity:

$$\begin{aligned} H_2 &= \int_V \boldsymbol{\Omega} \cdot \left[(\nabla \times \boldsymbol{B}) \times \frac{\partial \boldsymbol{B}}{\partial t} \right] \\ &= \int_V \frac{\partial \boldsymbol{B}}{\partial t} \cdot [\boldsymbol{\Omega} \times (\nabla \times \boldsymbol{B})] dV \, . \end{aligned} \tag{A6}$$

The third term in equation (A3) is first rewritten using a vector identity, and then Gauss's law is again applied:

$$\begin{aligned} H_3 &= -\int_V (\boldsymbol{\Omega} \cdot \boldsymbol{B}) \nabla \cdot \frac{\partial \boldsymbol{B}}{\partial t} \, dV \\ &= -\int_V \left\{ \nabla \cdot \left[(\boldsymbol{\Omega} \cdot \boldsymbol{B}) \frac{\partial \boldsymbol{B}}{\partial t} \right] - \frac{\partial \boldsymbol{B}}{\partial t} \cdot \nabla (\boldsymbol{\Omega} \cdot \boldsymbol{B}) \right\} dV \\ &= \int_S \frac{\partial \boldsymbol{B}}{\partial t} \cdot \hat{\boldsymbol{n}} (\boldsymbol{\Omega} \cdot \boldsymbol{B}) dS + \int_V \frac{\partial \boldsymbol{B}}{\partial t} \cdot \nabla (\boldsymbol{\Omega} \cdot \boldsymbol{B}) dV \, . \end{aligned} \tag{A7}$$

Finally, the fourth term in equation (A3) is already in the required form, and combining equations (A2), (A3), (A5), (A6), and (A7) establishes equations (6)–(9).

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